Period bound

Given target period φ, a retiming is feasible if it is legal and the clock period is at most φ

The idea

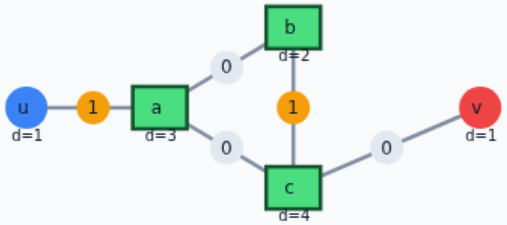
- Feas_check(data, 5, r) after forward on a: period 5 ≤ target 5 , passes

FEAS lite

PERIOD BOUND

FEAS lite

Step 1 / 5 · feas · module03-01-period-bound



Explanation

feas_check: legal AND period ≤ target.

- FEAS

```
toy
```

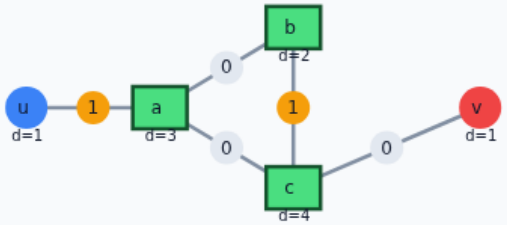
FEAS lite

Target 5

PERIOD BOUND

Target 5

Step 2 / 5 · target · module03-01-period-bound



Explanation

GOLDENS.periodTarget = 5.

- clock

```
T=5
```

Target 5

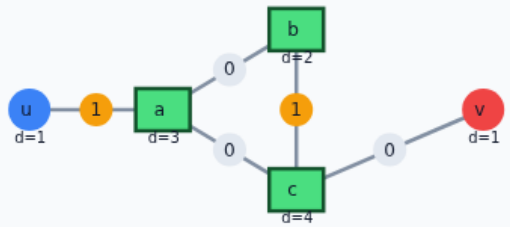
Baseline fails

PERIOD BOUND

Baseline fails

Step 3 / 5 · base-fail · module03-01-period-bound

P=8 feas=false



Explanation

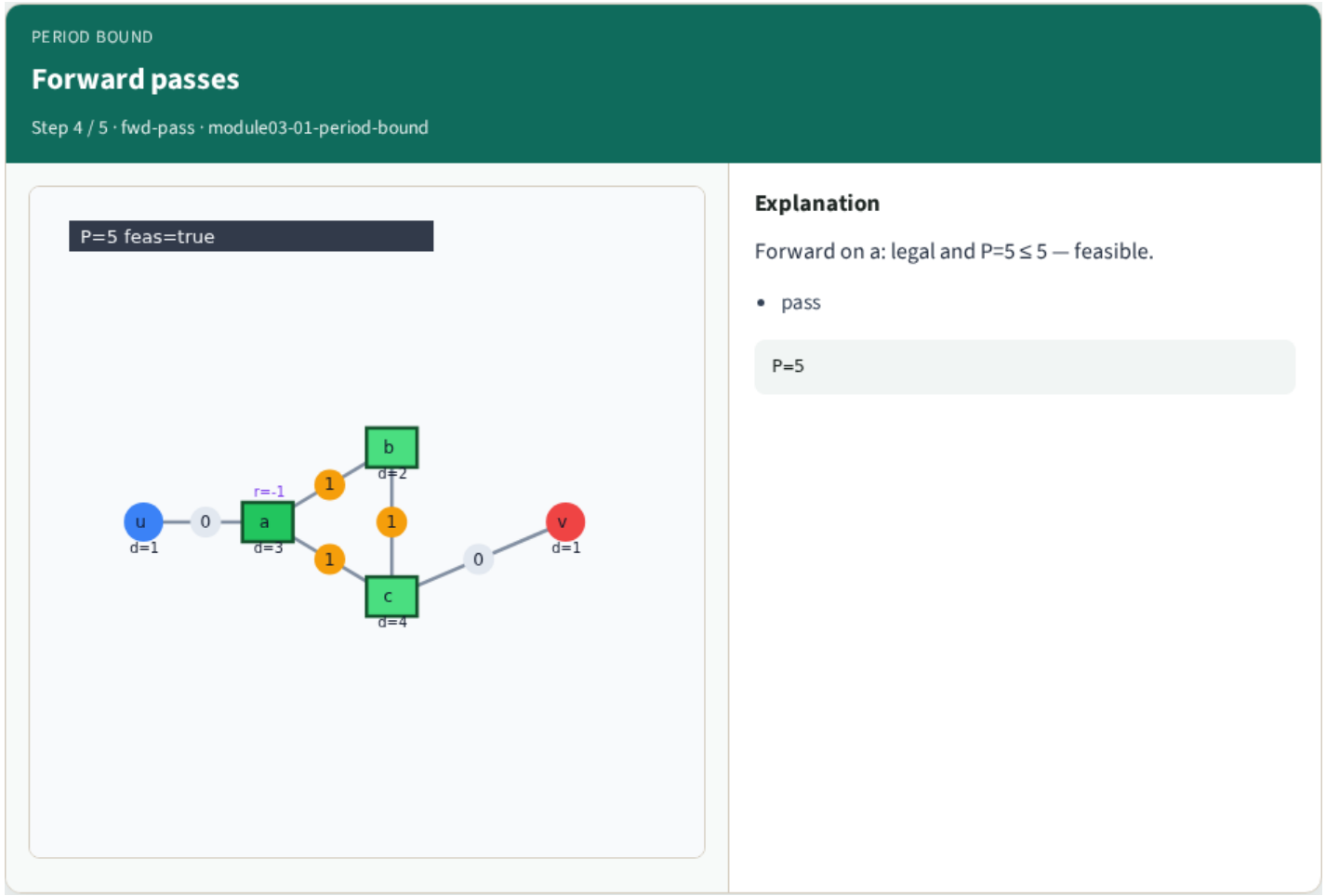
Period 8 > 5, so baseline is not feasible for the target.

- P=8

```
fail
```

Baseline fails

Forward passes



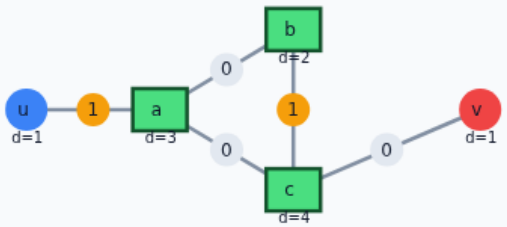
Forward passes

Full WD later

PERIOD BOUND

Full WD later

Step 5 / 5 · wd · module03-01-period-bound



Explanation

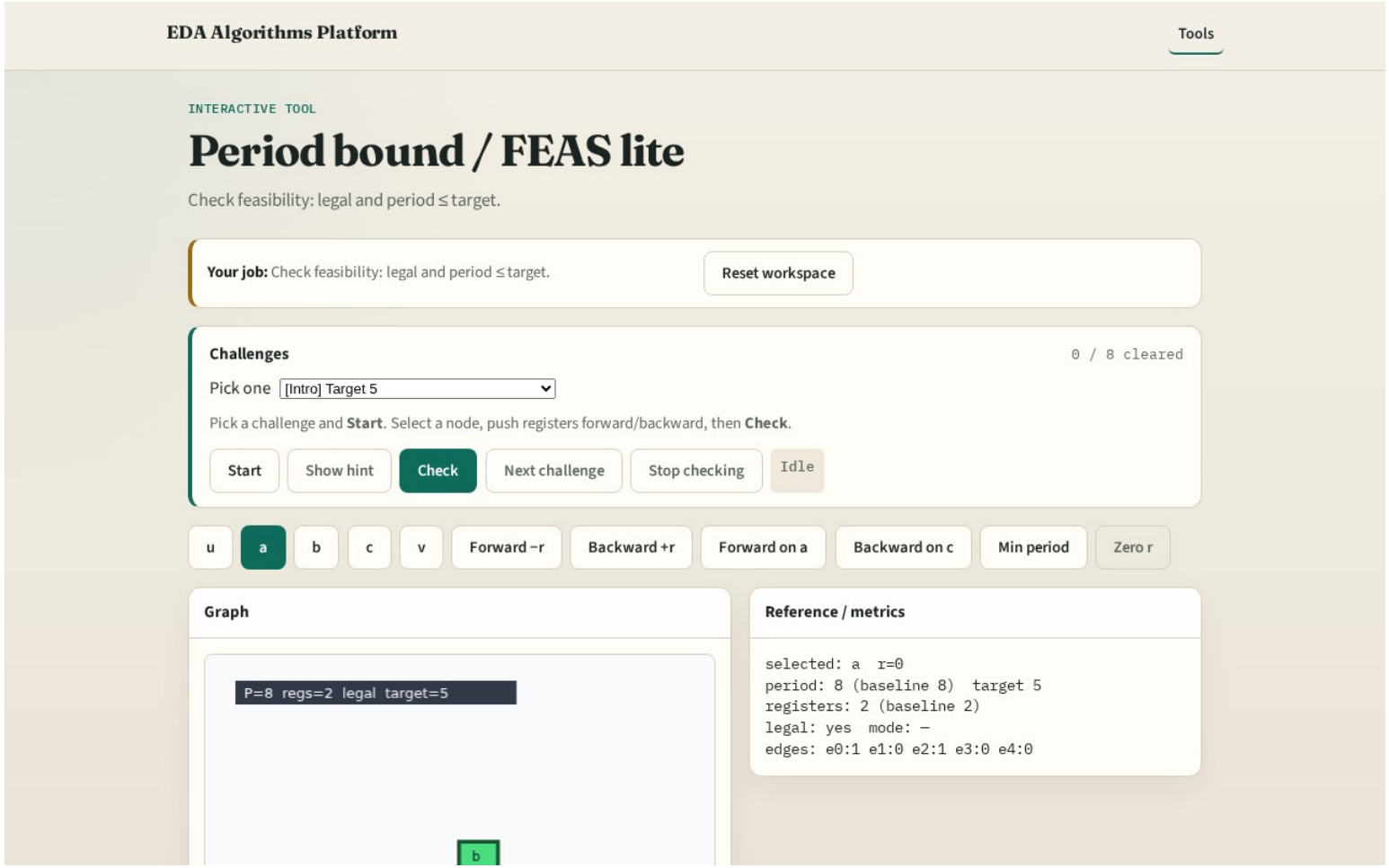
Real tools use W/D matrices; we only try a few retimings.

- literacy

```
not full LP
```

Full WD later

Browser lab track



Browser lab starter

Implement track

- Call feas_check(DATA, 5, r) with forward-on-a r; expect True

Pitfalls

- Forgetting legality in FEAS
- Comparing to target 8 instead of 5

Your turn

- Next: min-period search across candidate retimings